

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 7 EXERCISES

RELATIONAL ALGEBRA (RA)

BOATRESERVATIONS RELATIONAL DATABASE

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

Attribute names in italics are foreign key attributes.

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

11 tuples

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

5 tuples

EXERCISE 1

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor id.

👉 (22) (31) (74)

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

11 tuples

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

5 tuples

EXERCISE 1: SOLUTION

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor id.

👉 (22) (31) (74)

1. Is this a solution?

$\sigma_{\text{boatId}=103}(\pi_{\text{sailorId}}\text{Reserves})$

$\pi_{\text{sailorId}}\text{Reserves}$
sailorId
22
31
64
74
99

$\sigma_{\text{boatId}=103}$?
➔

2. Is this a solution?

$\pi_{\text{sailorId}}(\sigma_{\text{boatId}=103}\text{Reserves})$

$\sigma_{\text{boatId}=103}\text{Reserves}$		
sailorId	boatId	rDate
22	103	08-SEP-24
31	103	06-NOV-24
74	103	08-SEP-24

π_{sailorId}
➔

Query result

sailorId
22
31
74



EXERCISE 2

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio)

How to get the names of the sailors?

match sailorId in Reserves with sailorId in Sailor

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

11 tuples

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

5 tuples

EXERCISE 2

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

 (Dustin) (Lucy) (Horatio)

1. Is this a solution?

$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId} \wedge \text{boatId}=103}(\text{Reserves} \times \text{Sailor}))$

2. Is this a solution?

$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId}}((\sigma_{\text{boatId}=103}\text{Reserves}) \times \text{Sailor}))$

EXERCISE 2: SOLUTION I

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$$\pi_{sName}(\sigma_{Reserves.sailorId=Sailor.sailorId \wedge boatId=103}(Reserves \times Sailor))$$

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

11 tuples

X

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

=

How many tuples in the
Cartesian product?

11 x 13 = 143 tuples

EXERCISE 2: SOLUTION I (CONT'D)

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

Reserves X Sailor							
Reserves.sailorId	boatId	rDate	Sailor.sailorId	sName	hkid	rating	age
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
22	102	10-NOV24	99	Chris	H814243(7)	10	30
22	103	08-SEP-24	10	Emily	P936219(3)	null	16
22	103	08-SEP-24	15	Zoey	K091876(5)	null	17
22	103	08-SEP-24	22	Dustin	A298456(6)	7	45
22	103	08-SEP-24	29	Brutus	D746239(0)	1	33
22	103	08-SEP-24	31	Lucy	P930281(5)	10	55
22	103	08-SEP-24	32	Andy	M028391(4)	8	25
22	103	08-SEP-24	58	Rachael	A819204(3)	10	35
22	103	08-SEP-24	64	Horatio	F710943(2)	7	35
22	103	08-SEP-24	71	Zorba	P910562(4)	1	16
22	103	08-SEP-24	74	Horatio	T810463(1)	9	35
22	103	08-SEP-24	85	Amy	B719037(8)	3	25
22	103	08-SEP-24	95	Bob	E012735(5)	3	63
22	103	08-SEP-24	99	Chris	H814243(7)	10	30
22	104	10-OCT-24	10	Emily	P936219(3)	null	16
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

EXERCISE 2: SOLUTION I (CONT'D)

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$$\pi_{sName}(\sigma_{Reserves.sailorId=Sailor.sailorId \wedge boatId=103}(Reserves \times Sailor))$$

$\sigma_{Reserves.sailorId=Sailor.sailorId \wedge boatId=103}(Reserves \times Sailor)$							
Reserves.sailorId	boatId	rDate	Sailor.sailorId	sName	hkid	rating	age
22	103	08-SEP-24	22	Dustin	A298456(6)	7	45
31	103	06-NOV-24	31	Lucy	P930281(5)	10	55
74	103	08-SEP-24	74	Horatio	T810463(1)	9	35

Apply π_{sName} to above result:

Query result

sName
Dustin
Lucy
Horatio



EXERCISE 2: SOLUTION 2

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$$\pi_{sName}(\sigma_{Reserves.sailorId=Sailor.sailorId}((\sigma_{boatId=103}Reserves) \times Sailor))$$

$\sigma_{boatId=103}Reserves$		
sailorId	boatId	rDate
22	103	08-SEP-24
31	103	06-NOV-24
74	103	08-SEP-24

X

Sailor				
sailorId	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

=

How many tuples in the
Cartesian product?

3 x 13 = 39 tuples

13 tuples

EXERCISE 2: SOLUTION 2 (CONTD)

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$(\sigma_{\text{boatId}=103} \text{Reserves}) \times \text{Sailor}$							
Reserves.sailorId	boatId	rDate	Sailor.sailorId	sName	hkid	rating	age
22	103	08-SEP-24	10	Emily	P936219(3)	null	16
22	103	08-SEP-24	15	Zoey	K091876(5)	null	17
22	103	08-SEP-24	22	Dustin	A298456(6)	7	45
22	103	08-SEP-24	29	Brutus	D746239(0)	1	33
22	103	08-SEP-24	31	Lucy	P930281(5)	10	55
22	103	08-SEP-24	32	Andy	M028391(4)	8	25
22	103	08-SEP-24	58	Rachael	A819204(3)	10	17
22	103	08-SEP-24	64	Horatio	F710943(2)	7	35
22	103	08-SEP-24	71	Zorba	P910562(4)	1	16
22	103	08-SEP-24	74	Horatio	T810463(1)	9	35
22	103	08-SEP-24	85	Amy	B719037(8)	3	25
22	103	08-SEP-24	95	Bob	E012735(5)	3	63
22	103	08-SEP-24	99	Chris	H814243(7)	10	30
31	103	06-NOV-24	10	Emily	P936219(3)	null	16
31	103	06-NOV-24	15	Zoey	K091876(5)	null	17
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

EXERCISE 2: SOLUTION 2 (CONTD)

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio)

$\pi_{sName}(\sigma_{Reserves.sailorId=Sailor.sailorId}((\sigma_{boatId=103}Reserves) \times Sailor))$

$\sigma_{Reserves.sailorId=Sailor.sailorId}((\sigma_{boatId=103}Reserves) \times Sailor)$							
Reserves.sailorId	boatId	rDate	Sailor.sailorId	sName	hkid	rating	age
22	103	08-SEP-24	22	Dustin	A298456(6)	7	45
31	103	06-NOV-24	31	Lucy	P930281(5)	10	55
74	103	08-SEP-24	74	Horatio	T810463(1)	9	35

Apply π_{sName} to above result:

Query result

sName
Dustin
Lucy
Horatio



EXERCISE 2: SUMMARY

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

 (Dustin) (Lucy) (Horatio)

1. Is this a solution? ✓

$$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId} \wedge \text{boatId}=103}(\text{Reserves} \times \text{Sailor}))$$

Cartesian
product result:
143 tuples

2. Is this a solution? ✓

$$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId}}((\sigma_{\text{boatId}=103}\text{Reserves}) \times \text{Sailor}))$$

Cartesian
product result:
39 tuples

To be continued ...

EXERCISE 2: SOLUTION 3

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$$\pi_{sName}((\sigma_{boatId=103} Reserves) \text{ JOIN } Sailor)$$

$\sigma_{boatId=103} Reserves$		
sailorId	boatId	rDate
22	103	08-SEP-24
31	103	06-NOV-24
74	103	08-SEP-24

JOIN

Sailor				
sailorId	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

=

How many tuples in the join result?

3 tuples

13 tuples

EXERCISE 2: SOLUTION 3 (CONTD)

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

$$\pi_{sName}((\sigma_{boatId=103}Reserves) \text{ JOIN } Sailor)$$

$\sigma_{Reserves.sailorId=Sailor.sailorId}((\sigma_{boatId=103}Reserves) \times Sailor)$						
sailorId	boatId	rDate	sName	hkid	rating	age
22	103	08-SEP-24	Dustin	A298456(6)	7	45
31	103	06-NOV-24	Lucy	P930281(5)	10	55
74	103	08-SEP-24	Horatio	T810463(1)	9	35

Apply π_{sName} to above result:

Query result	
sName	
Dustin	
Lucy	
Horatio	



EXERCISE 2: SUMMARY

Find sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio)

☞ **All three queries get the correct result, BUT ...**

1. Is this a solution? ✓

$$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId} \wedge \text{boatId}=103}(\text{Reserves} \times \text{Sailor}))$$

Cartesian
product result:
143 tuples

2. Is this a solution? ✓

$$\pi_{\text{sName}}(\sigma_{\text{Reserves.sailorId}=\text{Sailor.sailorId}}((\sigma_{\text{boatId}=103}\text{Reserves}) \times \text{Sailor}))$$

Cartesian
product result:
33 tuples

3. Is this a solution? ✓

$$\pi_{\text{sName}}((\sigma_{\text{boatId}=103}\text{Reserves}) \text{ JOIN } \text{Sailor})$$

Join result:
3 tuples

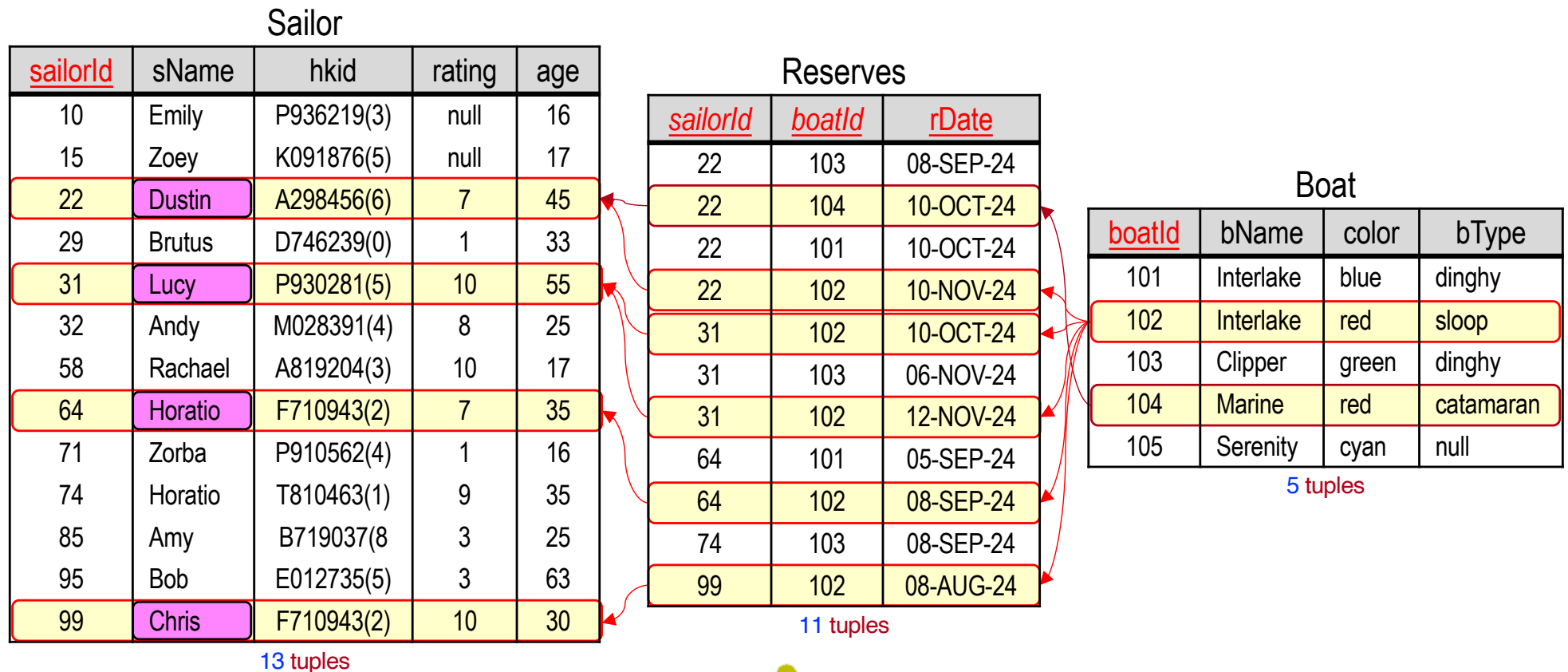
Query Optimization

Relational database systems do such optimizations based on relational algebra.

EXERCISE 3

Find sailors who have reserved a red boat.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio) (Chris)



EXERCISE 3: SOLUTION

Find sailors who have reserved a red boat.
Include in the query result tuples attribute sailor name.

$\pi_{sName}((\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor})$

$\sigma_{color='red'}Boat$			
boatId	bName	color	bType
102	Interlake	red	sloop
104	Marine	red	catamaran

JOIN

Reserves		
<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

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EXERCISE 3: SOLUTION (CONTD)

Find sailors who have reserved a red boat.
Include in the query result tuples attribute sailor name.

$\pi_{sName}((\sigma_{color='red'}Boat) \text{ JOIN } Reserves \text{ JOIN } Sailor)$

$(\sigma_{color='red'}Boat) \text{ JOIN } Reserves$					
<u>bName</u>	color	bType	sailorId	boatId	rDate
Interlake	red	sloop	22	102	10-NOV-24
Marine	red	catamaran	22	104	10-OCT-24
Interlake	red	sloop	31	102	10-NOV-24
Interlake	red	sloop	31	102	12-NOV-24
Interlake	red	sloop	64	102	08-SEP-24
Interlake	red	sloop	99	102	08-AUG-24

JOIN

Sailor				
<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	8	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	35
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	10	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	H814243(7)	10	30

=

13 tuples



EXERCISE 3: SOLUTION (CONTD)

Find sailors who have reserved a red boat.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio) (Chris)

$\pi_{sName}((\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor})$

$(\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor}$									
bName	color	bType	sailorId	boatId	rDate	sName	hkid	rating	age
Interlake	red	sloop	22	102	10-NOV-24	Dustin	A298456(6)	7	45
Marine	red	catamaran	22	104	10-OCT-24	Dustin	A298456(6)	7	45
Interlake	red	sloop	31	102	10-NOV-24	Lucy	P930281(5)	10	55
Interlake	red	sloop	31	102	12-NOV-24	Lucy	P930281(5)	10	55
Interlake	red	sloop	64	102	08-SEP-24	Horatio	F710943(2)	7	35
Interlake	red	sloop	99	102	08-AUG-24	Chris	H814243(7)	10	30

Apply π_{sName} to above result:

Query result

sName
Dustin
Lucy
Horatio
Chris



EXERCISE 4

Find boats that have never been reserved.
Include in the query result tuples attribute boat name.

☞ (Serenity)

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

11 tuples

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

5 tuples

EXERCISE 4: SOLUTION

Find boats that have never been reserved.
Include in the query result tuples attribute boat name.

 (Serenity)

$\pi_{bName}(\sigma_{sailorId=null}(Boat \bowtie Reserves))$

Boat				Reserves			Boat $\bowtie$ Reserves					
<u>boatId</u>	bName	color	bType	<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>	boatId	bName	color	bType	sailorId	rDate
101	Interlake	blue	dinghy	22	103	08-SEP-24	101	Interlake	blue	dinghy	22	10-OCT-24
102	Interlake	red	sloop	22	104	10-OCT-24	102	Interlake	red	sloop	22	10-NOV-24
103	Clipper	green	dinghy	22	101	10-OCT-24	103	Clipper	green	dinghy	22	08-SEP-24
104	Marine	red	catamaran	22	102	10-NOV-24	104	Marine	red	catamaran	22	10-OCT-24
105	Serenity	cyan	null	31	102	10-OCT-24	102	Interlake	red	sloop	31	10-OCT-24
				31	103	06-NOV-24	103	Clipper	green	dinghy	31	06-NOV-24
				31	102	12-NOV-24	102	Interlake	red	sloop	31	12-NOV-24
				64	101	05-SEP-24	101	Interlake	blue	dinghy	64	05-SEP-24
				64	102	08-SEP-24	102	Interlake	red	sloop	64	08-SEP-24
				74	103	08-SEP-24	103	Clipper	green	dinghy	74	08-SEP-24
				99	102	08-AUG-24	102	Interlake	red	sloop	99	08-AUG-24
							105	Serenity	cyan	null	null	null

EXERCISE 5: SOLUTION

Using set
difference on sName

Find sailors who have reserved a red boat, but not a green boat.
Include in the query result tuples attribute sailor name.

☞ (Horatio)
(Chris)

Is this a
solution?
No! Why?

$\pi_{sName}((\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor})$
-
 $\pi_{sName}((\sigma_{color='green'}Boat) \text{ JOIN Reserves JOIN Sailor})$

Since sName is not unique,
different sailors with the
same name (e.g., Horatio)
will be removed by the set
difference operation.

$(\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor}$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	H814243(7)	10	30

sailors who have reserved red boats

π_{sName}
→

sName
Dustin
Lucy
Horatio
Chris

How to
solve this
problem?

Query
result

-

→

sName
Chris

X

$(\sigma_{color='green'}Boat) \text{ JOIN Reserves JOIN Sailor}$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
103	Clipper	green	dinghy	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	dinghy	06-NOV-24	31	Lucy	P930281(5)	10	55
103	Clipper	green	dinghy	08-SEP-24	74	Horatio	T810463(1)	7	35

sailors who have reserved green boats

π_{sName}
→

sName
Dustin
Lucy
Horatio



EXERCISE 5: SOLUTION (CONTD)

Using set difference
on sName, sailorId

Find sailors who have reserved a red boat, but not a green boat.
Include in the query result tuples attribute sailor name.

☞ (Horatio)
(Chris)

Carry along an
attribute whose
values are unique.

$\pi_{sName, sailorId}((\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor})$

$\pi_{sName, sailorId}((\sigma_{color='green'}Boat) \text{ JOIN Reserves JOIN Sailor})$

Query result

sName	sailorId
Horatio	64
Chris	99



$(\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor}$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	H814243(7)	10	30

$\pi_{sName, sailorId}$

sName	sailorId
Dustin	22
Lucy	31
Horatio	64
Chris	99

sailors who have reserved a red boat

$(\sigma_{color='green'}Boat) \text{ JOIN Reserves JOIN Sailor}$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
103	Clipper	green	dinghy	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	dinghy	06-NOV-24	31	Lucy	P930281(5)	10	55
103	Clipper	green	dinghy	08-SEP-24	74	Horatio	T810463(1)	7	35

$\pi_{sName, sailorId}$

sName	sailorId
Dustin	22
Lucy	31
Horatio	74

sailors who have reserved a green boat



EXERCISE 6: SOLUTION I

Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

 (Dustin) (Lucy)

Is this a
solution?
No! Why?

$\pi_{sName}((\sigma_{color='red' \wedge color='green'} Boat) \text{ JOIN Reserves JOIN Sailor})$

Nothing is selected! Why?

The condition $color='red' \wedge color='green'$ can never be satisfied!

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

 (Dustin) (Lucy)

EXERCISE 6: SOLUTION 2

$\pi_{sName}((\sigma_{color='red' \vee color='green'} Boat) JOIN Reserves JOIN Sailor)$

Is this a solution?
No! Why?

$(\sigma_{color='red' \vee color='green'} Boat) JOIN Reserves JOIN Sailor$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	F710943(2)	10	30
103	Clipper	green	clipper	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	clipper	06-NOV-24	31	Lucy	P930281(5)	8	55
103	Clipper	green	clipper	08-SEP-24	74	Horatio	T810463(1)	7	35

sailors who have reserved a red or a green boat

π_{sName}


Query result

sName
Dustin
Lucy
Horatio
Chris

X

The condition $color='red' \vee color='green'$ includes sailors who have reserved only a red or only a green boat, as well as both a red and a green boat!

Must identify sailors who have reserved red boats, sailors who have reserved green boats, then find the sailors who are in both sets. $\Rightarrow$ intersection

Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

EXERCISE 6: SOLUTION 3

Intersect the join result

 (Dustin) (Lucy)

Is this a solution?
No! Why?

$$\pi_{sName}(((\sigma_{color='red'}Boat) \text{ JOIN Reserves JOIN Sailor}) \cap ((\sigma_{color='green'}Boat) \text{ JOIN Reserves JOIN Sailor}))$$

($\sigma_{color='red'}Boat$) JOIN Reserves JOIN Sailor									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	H814243(7)	10	30

sailors who have reserved a red boat

$\cap$

($\sigma_{color='green'}Boat$) JOIN Reserves JOIN Sailor									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
103	Clipper	green	dinghy	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	dinghy	06-NOV-24	31	Lucy	P930281(5)	10	55
103	Clipper	green	dinghy	08-SEP-24	74	Horatio	T810463(1)	7	35

sailors who have reserved a green boat



The query result is empty!



Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

EXERCISE 6: SOLUTION 4

Intersect
on sName

(Dustin) (Lucy)

Is this a
solution?
No! Why?

$$\pi_{sName}((\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor) \cap \pi_{sName}((\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor)$$

(Dustin) (Lucy)

Since **sName** is **not unique**, the intersection may contain incorrect tuples (i.e., Horatio is not unique).

$(\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	H814243(7)	10	30

sailors who have reserved a red boat

$(\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
103	Clipper	green	dinghy	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	dinghy	06-NOV-24	31	Lucy	P930281(5)	10	55
103	Clipper	green	dinghy	08-SEP-24	74	Horatio	T810463(1)	7	35

sailors who have reserved a green boat

π_{sName}

sName
Dustin
Lucy
Horatio
Chris

$\cap$

π_{sName}

sName
Dustin
Lucy
Horatio

X

How to
solve this
problem?

Query
result



Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy)

EXERCISE 6: SOLUTION 5

Intersect on
sName, sailorId.

Carry along an
attribute whose
values are unique.

$\pi_{sName, sailorId}((\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor)$

$\pi_{sName, sailorId}((\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor)$

☞ (Dustin) (Lucy)

$(\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
102	Interlake	red	sloop	10-NOV-24	22	Dustin	A298456(6)	7	45
104	Marine	red	catamaran	10-OCT-24	22	Dustin	A298456(6)	7	45
102	Interlake	red	sloop	10-OCT-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	12-NOV-24	31	Lucy	P930281(5)	10	55
102	Interlake	red	sloop	08-SEP-24	64	Horatio	F710943(2)	7	35
102	Interlake	red	sloop	08-AUG-24	99	Chris	H814243(7)	10	30

sailors who have reserved a red boat

$\pi_{sName, sailorId}$



sName	sailorId
Dustin	22
Lucy	31
Horatio	64
Chris	99

$\cap$



Query result

sName	sailorId
Dustin	22
Lucy	31



$(\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor$									
boatId	bName	color	bType	rDate	sailorId	sName	hkid	rating	age
103	Clipper	green	dinghy	08-SEP-24	22	Dustin	A298456(6)	7	45
103	Clipper	green	dinghy	06-NOV-24	31	Lucy	P930281(5)	10	55
103	Clipper	green	dinghy	08-SEP-24	74	Horatio	T810463(1)	7	35

sailors who have reserved a green boat

$\pi_{sName, sailorId}$



sName	sailorId
Dustin	22
Lucy	31
Horatio	74



Find sailors who have reserved both a red and a green boat.
Include in the query result tuples attribute sailor name.

EXERCISE 6: SOLUTION 6

Join on sailorId

 (Dustin) (Lucy)

Is this a solution?

Yes!

$\pi_{sName, sailorId}((\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor)$
 $\pi_{sName, sailorId}((\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor)$

sailors who have reserved a red boat

$\pi_{sName, sailorId}((\sigma_{color='red'}Boat) JOIN Reserves JOIN Sailor)$

sName	sailorId
Dustin	22
Lucy	31
Horatio	64
Chris	99

JOIN_{sailorId}

$\pi_{sName, sailorId}((\sigma_{color='green'}Boat) JOIN Reserves JOIN Sailor)$

sName	sailorId
Dustin	22
Lucy	31
Horatio	74

sailors who have reserved a green boat

$\pi_{sName, sailorId}$ 

Query result

sName	sailorId
Dustin	22
Lucy	31



EXERCISE 7: SOLUTION

Find boats reserved more than once by the same sailor.
Include in the query result tuples attribute boat id.

 (102)

We need to use rename: $\rho_{R1}(\text{Reserves})$, $\rho_{R2}(\text{Reserves})$

$$\pi_{R1.boatId}(\sigma_{R1.boatId=R2.boatId \wedge R1.sailorId=R2.sailorId \wedge R1.rDate <> R2.rDate}(\rho_{R1}(\text{Reserves}) \times \rho_{R2}(\text{Reserves})))$$

Or equivalently:

$$\pi_{R1.boatId}(\rho_{R1}(\text{Reserves}) \text{ JOIN }_{R1.boatId=R2.boatId \wedge R1.sailorId=R2.sailorId \wedge R1.rDate <> R2.rDate} \rho_{R2}(\text{Reserves}))$$

EXERCISE 7: SOLUTION (CONTD)

Find boats reserved more than once by the same sailor.
Include in the query result tuples attribute boat id.

 (102)

$\pi_{R1.boatId}(\sigma_{R1.boatId=R2.boatId \wedge R1.sailorId=R2.sailorId \wedge R1.rDate <> R2.rDate} (R1 \times R2))$

R1

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

X

R2

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-24
22	104	10-OCT-24
22	101	10-OCT-24
22	102	10-NOV-24
31	102	10-OCT-24
31	103	06-NOV-24
31	102	12-NOV-24
64	101	05-SEP-24
64	102	08-SEP-24
74	103	08-SEP-24
99	102	08-AUG-24

=

EXERCISE 7: SOLUTION (CONTD)

Find boats reserved more than once by the same sailor.
Include in the query result tuples attribute boat id.

 (102)

$\pi_{R1.boatId}(\sigma_{R1.boatId=R2.boatId \wedge R1.sailorId=R2.sailorId \wedge R1.rDate <> R2.rDate} (R1 \times R2))$

$\sigma_{R1.sailorId=R2.sailorId \wedge R1.boatId=R2.boatId \wedge R1.rDate <> R2.rDate}$					
R1.sailorId	R1.boatId	R1.rDate	R2.sailorId	R2.boatId	R2.rDate
31	102	10-OCT-24	31	102	10-OCT-24
31	102	12-NOV-24	31	102	12-NOV-24

$\pi_{R1.boatId} =$

Query result

boatId
102

EXERCISE 7: SOLUTION (CONTD)

What do we get if we omit $R1.rDate \neq R2.rDate$?

$\sigma_{R1.boatId=R2.boatId \wedge R1.sailorId=R2.sailorId}$					
R1.boatId	R1.sailorId	R1.rDate	R2.boatId	R2.sailorId	R2.rDate
101	22	10-OCT-24	101	22	10-OCT-24
101	64	05-SEP-24	101	64	05-SEP-24
102	22	10-NOV-24	102	22	10-NOV-24
102	31	10-OCT-24	102	31	12-NOV-24
102	31	12-NOV-24	102	31	12-NOV-24
102	31	12-NOV-14	103	31	10-OCT-24
102	31	10-OCT-24	102	31	10-OCT-24
102	64	08-SEP-24	102	64	08-SEP-24
102	99	08-AUG-24	102	99	08-AUG-24
103	22	08-SEP-24	103	22	08-SEP-24
103	31	06-NOV-24	103	31	06-NOV-24
103	74	08-SEP-24	103	74	08-SEP-24
104	22	10-OCT-24	104	22	10-OCT-24

$\pi_{R1.boatId} =$

Query result

boatId
101
102
103
104

Boats that have been reserved at least once.

EXERCISE 7: SOLUTION (CONTD)

What do we get if we omit $R1.sailorId=R2.sailorId$?

$\sigma_{R1.boatId=R2.boatId \wedge R1.rDate <> R2.rDate}$					
R1.boatId	R1.sailorId	R1.rDate	R2.boatId	R2.sailorId	R2.rDate
101	22	10-OCT-24	101	64	05-SEP-24
101	64	05-SEP-24	101	22	10-OCT-24
102	22	10-NOV-24	102	99	08-AUG-24
102	22	10-NOV-24	102	31	10-OCT-24
102	22	10-NOV-24	102	64	08-SEP-24
102	22	10-NOV-24	102	31	12-NOV-24
102	31	12-NOV-24	102	22	10-NOV-24
102	31	10-OCT-24	102	22	10-NOV-24
102	31	12-NOV-24	102	64	08-SEP-24
⋮	⋮	⋮	⋮	⋮	⋮
102	31	12-NOV-24	102	31	10-OCT-24
102	64	08-SEP-24	102	99	08-AUG-24
102	64	08-SEP-24	102	31	12-NOV-24
102	64	08-SEP-24	102	31	10-OCT-24
102	64	08-SEP-24	102	22	10-NOV-24
103	22	08-SEP-24	103	31	06-NOV-24
103	31	06-NOV-24	103	74	08-SEP-24
103	31	06-NOV-24	103	22	08-SEP-24
103	74	08-SEP-24	103	31	06-NOV-24

$\pi_{R1.boatId} =$

Query result

boatId
101
102
103

Boats that have been reserved more than once.